

GIS-I: GEOGRAPHIC INFORMATION SYSTEMS

CHAPTER THREE

SPATIAL QUERY

BSc in Surveying Engineering

Lecture Note and Practical Guide

3.1 Learning Objectives

After completing this chapter, students should be able to:

1. Define the concept of spatial query.
2. Explain the difference between attribute queries and spatial queries.
3. Perform simple attribute queries.
4. Explain common spatial relationships such as within, contains, intersects, overlaps, touches, crosses, disjoint, and near.
5. Perform spatial selection using GIS software.
6. Combine spatial and attribute conditions.
7. Use SQL expressions for attribute queries.
8. Apply spatial queries to surveying and GIS problems.
9. Interpret and present the results of spatial queries.
10. Distinguish between selection, spatial query, and spatial analysis.

3.2 Introduction

One of the most important capabilities of a Geographic Information System (GIS) is the ability to ask questions about geographic data and obtain answers based on location, attributes, or both. This process is called a spatial query.

In a traditional database, we may ask, “Which students are older than 25 years?” In GIS, we can ask questions such as: “Which buildings are located within 500 meters of a road?”, “Which parcels intersect the proposed road?”, or “Which villages are located inside the flood-prone area?”

Spatial query therefore allows GIS users to retrieve specific features from a large dataset based on where the features are located, what their attributes are, or the relationship between different geographic features.

3.3 What is a Query?

A query is a question or request made to a database in order to retrieve particular information.

Suppose we have a GIS layer containing 5,000 buildings. Each building has information such as building type, population, area, and elevation. We could ask: “Show me all buildings with an area greater than 1,000 m².” The GIS selects the features satisfying that condition. This is an attribute query because the question is based on information stored in the attribute table.

Building ID	Type	Population	Area (m ²)	Elevation (m)
B001	Residential	6	120	1450
B002	School	350	2500	1460
B003	Residential	4	90	1455
B004	Hospital	120	4200	1470
B005	Office	25	850	1465

3.4 What is Spatial Query?

A spatial query is a query that retrieves geographic features according to their spatial location or spatial relationship with other geographic features.

In simple terms, a spatial query asks a GIS question about where something is, what it is near, what it intersects, what it contains, or how it relates spatially to another feature.

- Which buildings are within 500 m of a road?
- Which parcels intersect the project boundary?
- Which villages are inside the administrative district?
- Which roads cross the river?
- Which schools are nearest to residential areas?
- Which buildings are outside the flood zone?

3.5 Spatial Query vs. Attribute Query

Query Type	Basis	Example
Attribute Query	Information stored in the attribute table	Select parcels where Area > 5 hectares.
Spatial Query	Location or spatial relationship	Select parcels that intersect the proposed highway.
Combined Query	Attribute + spatial condition	Select residential buildings within 500 m of a major road.

3.6 Types of Spatial Queries

The major spatial relationships include:

- Within
- Contains
- Intersects
- Overlaps
- Touches
- Crosses
- Disjoint
- Near/Distance
- Within a specified distance
- Nearest feature

3.7 Query Using “Within”

The within relationship identifies features that are completely located inside another geographic feature.

Examples:

- Which villages are within District A?
- Which buildings are within the project boundary?
- Which settlements are within a watershed?

3.8 Query Using “Contains”

Contains is the reverse spatial relationship of within. If Village A is within District B, then District B contains Village A.

Examples:

- Which administrative district contains this GPS point?
- Which cadastral parcel contains this survey point?

3.9 Query Using “Intersects”

Two geographic features intersect when they have a spatial relationship involving a common location or area.

Examples:

- Which parcels intersect the proposed highway?
- Which rivers intersect the project boundary?
- Which roads intersect the study area?

3.10 Query Using “Overlaps”

Two geographic features overlap when they occupy some common geographic area, especially polygon features.

Examples:

- Which agricultural areas overlap the irrigation project area?
- Which land-use areas overlap the proposed development zone?

3.11 Query Using “Touches”

Two features touch when they meet at their boundary without necessarily overlapping.

Examples:

- Which parcels touch parcel P001?
- Which administrative units share a boundary?

3.12 Query Using “Crosses”

The crosses relationship is commonly used when a line passes through another geographic feature.

Examples:

- Which roads cross the river?
- Which pipelines cross roads?
- Which power lines cross the highway?

3.13 Query Using “Disjoint”

Two features are disjoint when they have no common spatial location.

Examples:

- Which villages are completely outside the flood-risk zone?
- Which parcels do not intersect the project boundary?

3.14 Distance-Based Spatial Query

A distance-based spatial query identifies features according to a specified distance from another feature.

Examples:

- Which houses are within 500 m of a road?
- Which villages are within 3 km of a health center?
- Which water points are within 1 km of settlements?

3.15 Nearest Feature Query

A nearest-feature query identifies the geographic feature located closest to another feature.

Examples:

- Which school is nearest to this residential area?
- Which road is nearest to this building?
- Which health center is closest to this village?

3.16 Attribute Query

An attribute query uses information stored in an attribute table. Common comparison and logical operators include:

Operator	Meaning
=	Equal to
<>	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to
AND	Both conditions must be true
OR	At least one condition must be true
NOT	Excludes a condition
LIKE	Pattern matching

3.17 Example of Attribute Query

Parcel_ID	LandUse	Area_ha	Value
P01	Residential	0.5	100000
P02	Agricultural	5.2	250000
P03	Residential	1.8	400000
P04	Forest	8.5	150000
P05	Agricultural	2.5	180000

To select agricultural parcels:

```
LandUse = 'Agricultural'
```

To select agricultural parcels larger than 3 hectares:

```
LandUse = 'Agricultural' AND Area_ha > 3
```

To select residential or forest parcels:

```
LandUse = 'Residential' OR LandUse = 'Forest'
```

3.18 Combining Attribute and Spatial Queries

The real power of GIS comes from combining spatial and attribute conditions. For example, to identify residential buildings within 500 m of a major road, two conditions are required:

```
Building_Type = 'Residential'
```

```
Distance from major road ≤ 500 m
```

The final result contains only buildings satisfying both conditions.

3.19 Spatial Query Example for Health Facility Planning

Suppose a local government wants to identify communities for a new health center. Available GIS layers include roads, villages, rivers, schools, existing health centers, administrative boundaries, and land-use maps.

- Which villages are within 2 km of a major road?
- Which villages are more than 5 km from an existing health center?
- Which suitable areas are outside the flood-risk zone?
- Which available land parcels are larger than 1 hectare?

Several queries can be combined to identify areas satisfying multiple requirements.

3.20 Spatial Query in Surveying Engineering

3.20.1 Cadastral Surveying

A surveyor may need to determine which parcels are affected by a proposed road. The road layer is queried against the cadastral parcel layer. The selected parcels can then support land-acquisition and cadastral assessment.

3.20.2 Route Surveying

- Which buildings are within 50 m of the road?
- Which rivers are crossed?
- Which parcels are affected?
- Which settlements are within 1 km?
- Which environmentally sensitive areas are intersected?

3.20.3 Topographic Surveying

- Elevation points above a particular elevation
- Areas within a specified slope range
- Drainage channels crossing a project boundary
- Terrain features within a construction site

For example, selecting areas where slope is greater than 30% can be treated as a raster or attribute-based query depending on how the slope information is stored.

3.21 Spatial Query with Raster Data

Spatial queries are not limited to vector data. Raster data can also be queried. A DEM, for example, stores elevation values in cells.

```
Elevation > 1200
```

The output can be a new raster identifying cells whose elevation satisfies the condition.

3.21.1 Raster Query Using NDVI

For an NDVI raster, a user may identify areas with relatively high vegetation values using a threshold such as:

```
NDVI > 0.5
```

Such queries can support vegetation monitoring, agricultural assessment, land degradation studies, forest monitoring, and environmental analysis.

3.22 Spatial Query Using Multiple Conditions

GIS allows several conditions to be combined. For example, suitable land may be defined as areas where slope is less than 15%, elevation is below 1,800 m, and the area is more than 500 m from a river.

```
Slope < 15%
```

```
AND Elevation < 1800 m
```

```
AND Distance from River > 500 m
```

This is an example of a multi-criteria spatial query. The result may be used as an input to a broader site suitability analysis.

3.23 Spatial Query and GIS Selection

A query often produces a selection. Selected features can be displayed, exported, calculated, symbolized, used in further analysis, or reported.

```
All Features → Apply Query → Selected Features → Display / Export / Analyze
```

3.24 Spatial Query in ArcGIS Desktop 10.8

In ArcGIS Desktop 10.8, spatial queries can commonly be performed using Select By Location.

General procedure:

11. Add the target and reference/source layers to ArcMap.
12. Open Selection → Select By Location.
13. Choose the target layer whose features will be selected.
14. Choose the source/reference layer.
15. Select the appropriate spatial relationship, such as intersect, within, contain, or within a distance of.
16. If required, specify the search distance and units.
17. Run the selection and inspect the selected features.

Example:

- Target layer: Buildings
- Source layer: Major Roads
- Spatial relationship: Within a distance of
- Distance: 500 meters

Result: buildings located within 500 m of the selected major roads.

3.25 Select By Attributes in ArcGIS Desktop 10.8

For attribute-based queries, use Selection → Select By Attributes.

Example: select residential land-use features:

```
"LANDUSE" = 'Residential'
```

Example: select areas larger than 5 hectares:

```
"AREA_HA" > 5
```

Example: combine two conditions:

```
"LANDUSE" = 'Residential' AND "AREA_HA" > 5
```

The exact SQL syntax can vary depending on the data source and field type.

3.26 Example: Querying Cadastral Parcels

Consider a cadastral layer containing Parcel_ID, Owner, LandUse, Area, and geometry, together with a proposed road layer.

Question: Which residential parcels larger than 1 hectare intersect the proposed road?

Attribute condition:

```
LandUse = 'Residential' AND Area > 1
```

Spatial condition:

```
Parcel INTERSECTS Proposed Road
```

The final selected parcels satisfy both requirements. This type of query is useful in land acquisition and infrastructure development.

3.27 Practical Example: Tepi Area

Consider a GIS project for an area around Tepi. Available datasets include roads, buildings, rivers, administrative boundaries, land-use/land-cover, DEM, health centers, and water points.

Question: Which residential buildings are located within 300 m of a major road?

18. Select residential buildings using the Building_Type field.
19. Select or identify major roads.
20. Apply a spatial query using the relationship Within 300 m of.
21. Review the selected buildings on the map and in the attribute table.
22. Export the selected features if a separate output layer is required.

Interpretation: The selected buildings represent residential structures with relatively close spatial access to the major road network. The result can support transportation accessibility studies, service planning, and infrastructure assessment.

3.28 Spatial Query Workflow

23. Define the geographic question clearly.
24. Identify the required datasets.
25. Check the coordinate reference systems.
26. Define the spatial relationship or attribute condition.
27. Apply the query.
28. Examine the selected features.
29. Export the result if necessary.
30. Interpret and communicate the result.

For distance-based analysis, an appropriate projected coordinate system is generally preferable because distance calculations depend on the coordinate reference system and its units.

3.29 Spatial Query vs. Spatial Analysis

Concept	Main Purpose	Example
Spatial Query	Find or select features satisfying a condition.	Which buildings are within 500 m of the road?
Spatial Analysis	Analyze geographic relationships to create information or support decisions.	Buffer, overlay, interpolation, density, network, terrain, or suitability analysis.

A spatial query can therefore be an important component of a larger spatial analysis workflow.

3.30 Advantages of Spatial Query

- Fast information retrieval from large spatial databases.
- Location-based decision support.
- Integration of multiple GIS datasets.
- Support for infrastructure, environmental, cadastral, and urban planning.
- Reproducible workflows that can be repeated when data are updated.

- Reduction of manual map inspection and feature selection.

3.31 Limitations and Important Considerations

- Data quality: inaccurate or incomplete spatial data can produce incorrect results.
- Coordinate system: inappropriate coordinate reference systems can affect distance calculations.
- Spatial accuracy: positional accuracy affects the reliability of selected features.
- Scale: datasets created at inappropriate scales may not be suitable for detailed engineering applications.
- Data currency: old datasets may not represent current geographic conditions.
- Query definition: an incorrectly defined spatial relationship may produce an inappropriate result.

3.32 Key Spatial Query Relationships

Relationship	Basic Meaning	Example
Within	A is inside B	Building within city
Contains	B contains A	City contains building
Intersects	Features have a spatial intersection	Road intersects parcel
Overlaps	Areas share some common area	Two land-use zones overlap
Touches	Features meet at a boundary	Adjacent parcels touch
Crosses	One feature passes across another	Road crosses river
Disjoint	No spatial intersection	Village outside flood zone
Near	Features are close	School near road
Within distance	Feature lies inside specified distance	Building within 500 m of road
Nearest	Identifies closest feature	Nearest health center

3.33 Chapter Summary

A spatial query is a fundamental GIS operation used to retrieve geographic features according to their location, spatial relationship, distance, or a combination of spatial and attribute conditions.

The most common relationships are within, contains, intersects, overlaps, touches, crosses, disjoint, near, within a specified distance, and nearest. Attribute queries use information stored in attribute tables, while spatial queries use geographic relationships. In practical GIS work, the greatest value often comes from combining both.

For Surveying Engineering, spatial queries are particularly useful in cadastral surveying, route surveying, urban planning, land administration, infrastructure planning, site selection, environmental assessment, transportation planning, facility accessibility, and raster-based analysis.

The basic principle is: Ask a geographic question → identify the required data → define the spatial relationship → apply the query → examine the selected features → interpret the result.

3.34 Review Questions

Short-answer questions:

31. What is a GIS query?

32. Define spatial query.
33. What is the difference between spatial query and attribute query?
34. Explain the meaning of within.
35. Explain the meaning of intersects.
36. What is the difference between contains and within?
37. What is a distance-based spatial query?
38. What is the purpose of a buffer in spatial querying?
39. What is a nearest-feature query?
40. Why is coordinate system important in distance-based spatial queries?

Discussion questions:

41. Explain how spatial queries can support cadastral surveying.
42. Discuss applications of spatial queries in route surveying.
43. Explain how spatial queries can be used for health-center site selection.
44. Discuss the difference between spatial query and spatial analysis.
45. Explain why data quality affects the reliability of spatial query results.

3.35 Practical Exercise: Spatial Query for Infrastructure Planning

Objective: To practice attribute and spatial queries using GIS software.

Data required:

- Building layer
- Road layer
- Parcel layer
- Administrative boundary
- Health-center layer

Tasks:

46. Select all residential buildings.
47. Select all parcels with an area greater than 1 hectare.
48. Select all buildings within 500 m of major roads.
49. Select all parcels intersecting the proposed road.
50. Identify residential parcels larger than 1 hectare that intersect the proposed road.
51. Identify the nearest health center to each village or settlement.

Expected output:

52. A map showing selected features.
53. An attribute table containing the selected features.
54. A short explanation of the query used.
55. A brief interpretation of the result.

3.36 Key Terms

Term	Meaning
Query	A request for information from a database.

Spatial Query	A query based on geographic location or spatial relationship.
Attribute Query	A query based on non-spatial attribute values.
Within	A feature is located inside another feature.
Contains	A feature spatially contains another feature.
Intersect	Two features have a common spatial relationship.
Overlap	Two areas share part of their spatial extent.
Buffer	A zone created around a feature at a specified distance.
Nearest	The closest feature according to a distance measure.
Select By Location	A GIS operation that selects features based on spatial relationships.
Select By Attributes	A GIS operation that selects features using attribute conditions.

3.37 References and Recommended Reading

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